

Project Title : Data Visualization Dashboard

Project Name : Sample Superstore Profitability Dashboard

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Internship : CodeAlpha Data Analytics Internship

Task : Task 3 – Data Visualization

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1. Business Problem

The company generates strong sales across multiple product categories, but profitability is not consistent. The objective of this dashboard is to present business performance through clear visualizations so that management can quickly identify profitable and loss-making areas and support data-driven decision-making.

2. Dashboard Objectives

- Design an executive-level business dashboard.
- Present important KPIs for quick business understanding.
- Visualize profitability across categories, regions, customer segments, discounts and shipping modes.
- Identify business areas requiring immediate attention.
- Support decision-making using data visualization.

3. Dataset Overview

The Sample Superstore dataset from Kaggle was used for this project. It belongs to the retail sales domain and contains 9,977 cleaned records with 13 features including Sales, Profit, Discount, Category, Sub-Category, Region, Segment and Ship Mode.

4. Data Cleaning

Data Quality Check	Observation	Action Taken
Missing Values	No missing values	No action required
Duplicate Records	17 duplicate records	Removed duplicates
Data Types	Appropriate	No action required

5. Dashboard Analysis

5.1 KPI Dashboard

Business Question: What is the overall business performance?

Observation: The KPI cards summarize overall business performance. Total Sales exceeded ₹2.29M while the profit margin remained 12.47%, indicating opportunities to improve profitability.



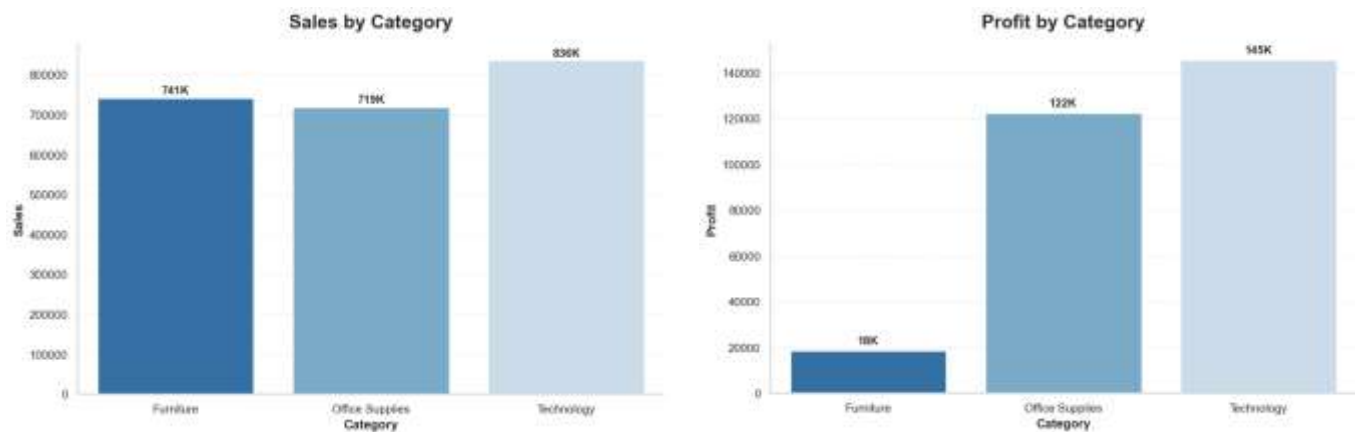
Key Findings: Total Sales ₹2.30M, Total Profit ₹0.29M, Profit Margin 12.47%, Total Orders 9,977, Average Discount 15.63%.

Business Insight: Monitoring profit margin along with sales provides a better picture of business performance.

5.2 Category Performance Analysis

Business Question: Which category contributes the highest sales and profit?

Observation: Technology generated the highest sales and profit. Furniture generated high sales but comparatively low profitability, while Office Supplies maintained better profit than Furniture despite lower sales.



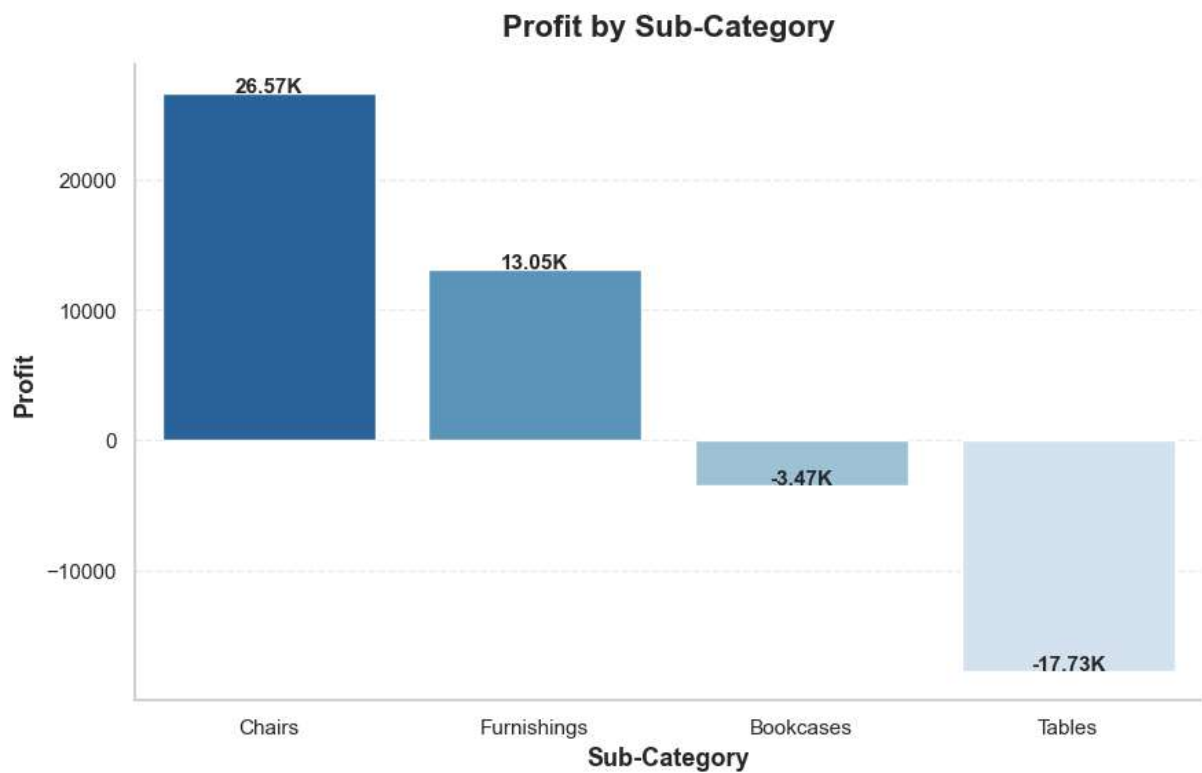
Key Findings: Furniture requires business attention because high sales are not translating into high profit.

Business Insight: Improving Furniture margins can significantly improve overall profitability.

5.3 Furniture Sub-Category Performance Analysis

Business Question: Which furniture sub-category is responsible for lower profit?

Observation: Tables generated high sales but recorded the largest losses. Chairs performed well, while Bookcases also showed negative profit.



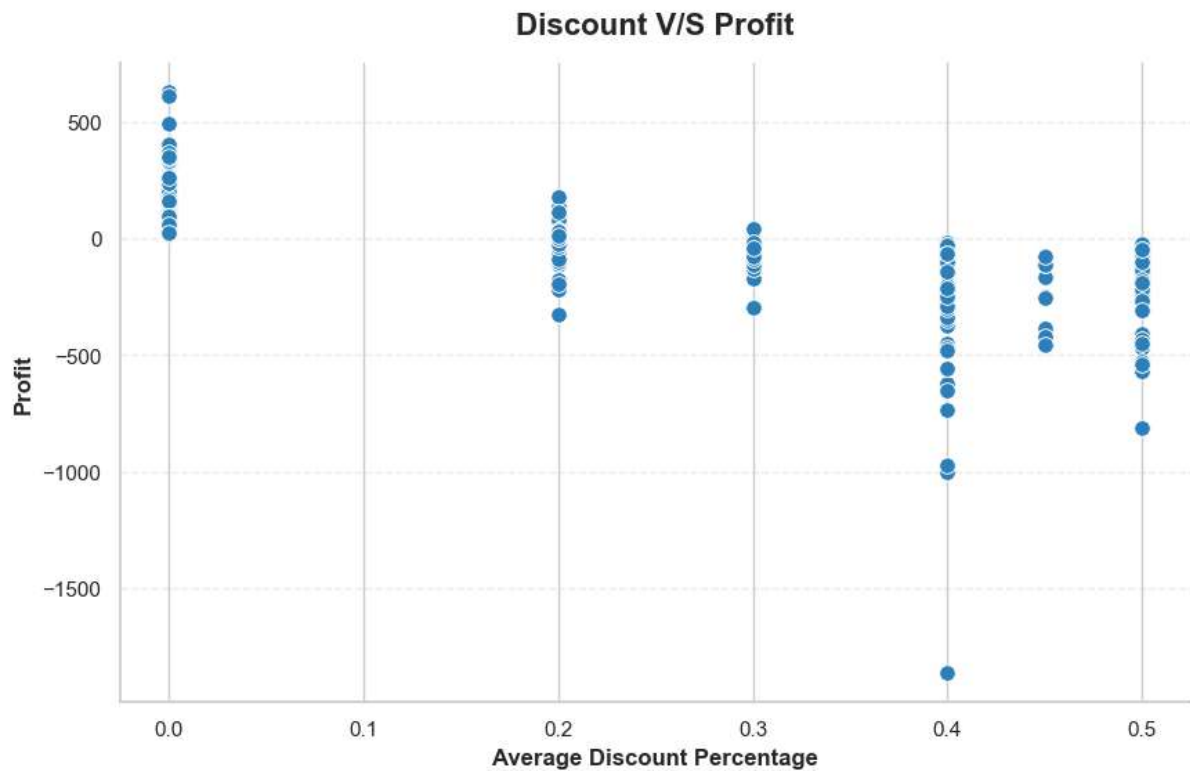
Key Findings: Tables are the major loss-making sub-category.

Business Insight: Pricing and discount strategy for Tables should be reviewed.

5.4 Discount Impact Analysis

Business Question: Do higher discounts reduce profitability?

Observation: The scatter plot indicates a negative relationship between discount and profit for the Tables sub-category. Higher discounts are associated with larger losses.



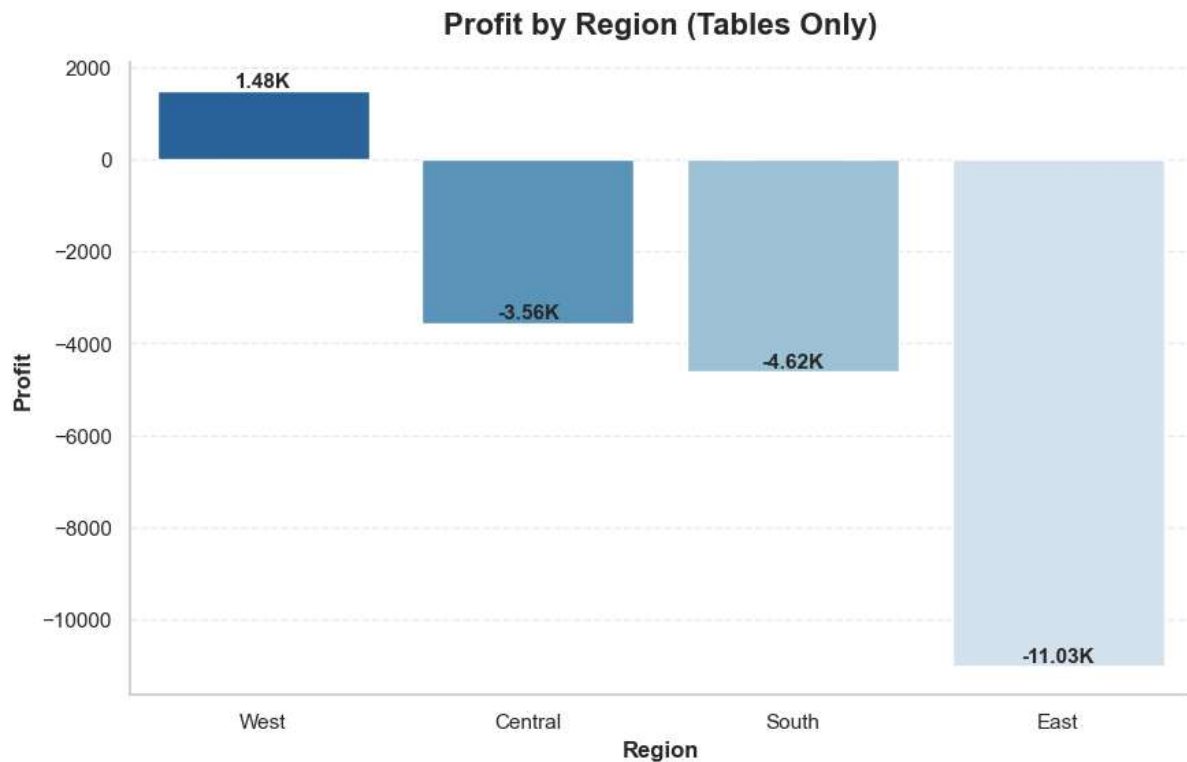
Key Findings: Higher discounts generally reduce profitability.

Business Insight: Discount policies should be optimized to balance sales and profit.

5.5 Regional Performance Analysis

Business Question: Which region performs the best and worst?

Observation: The West region generated the highest profit whereas the East region recorded the lowest profitability.



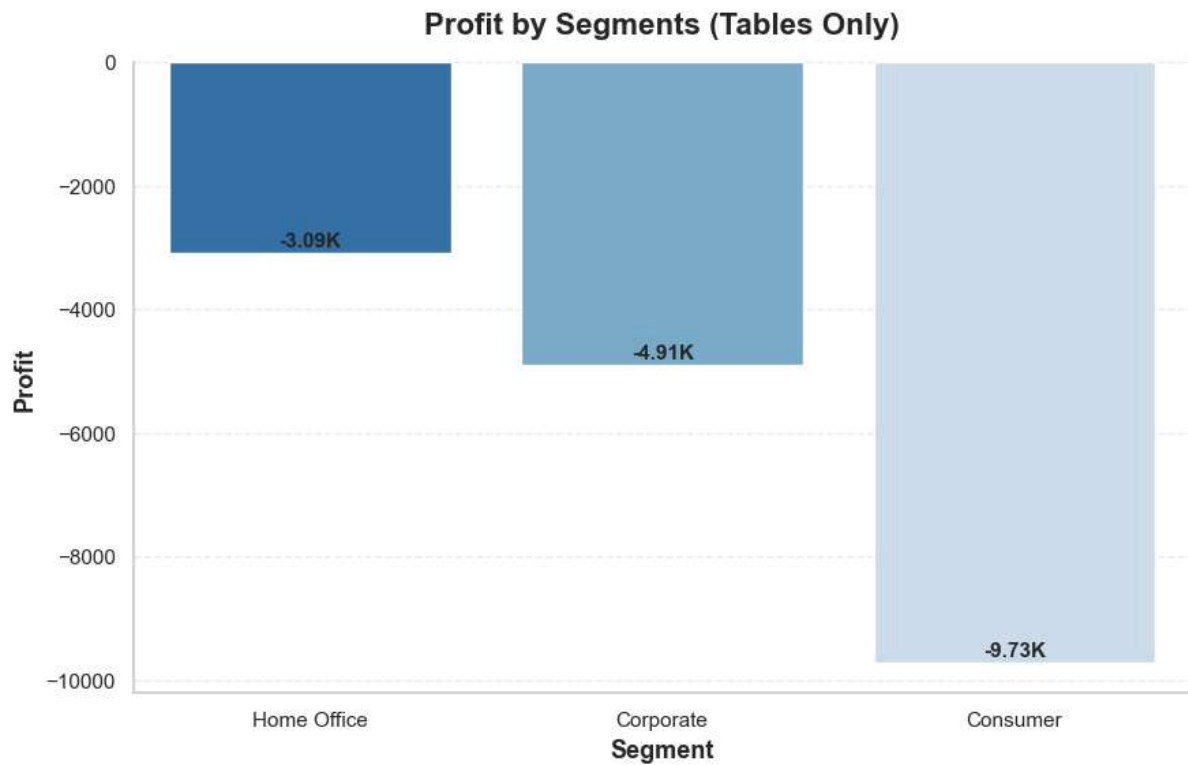
Key Findings: Profitability varies across regions.

Business Insight: Investigate pricing, demand and operational factors in the East region.

5.6 Customer Segment Performance Analysis

Business Question: Which customer segment contributes the highest profitability?

Observation: The Consumer segment generated the highest profit while the Home Office segment generated the lowest.



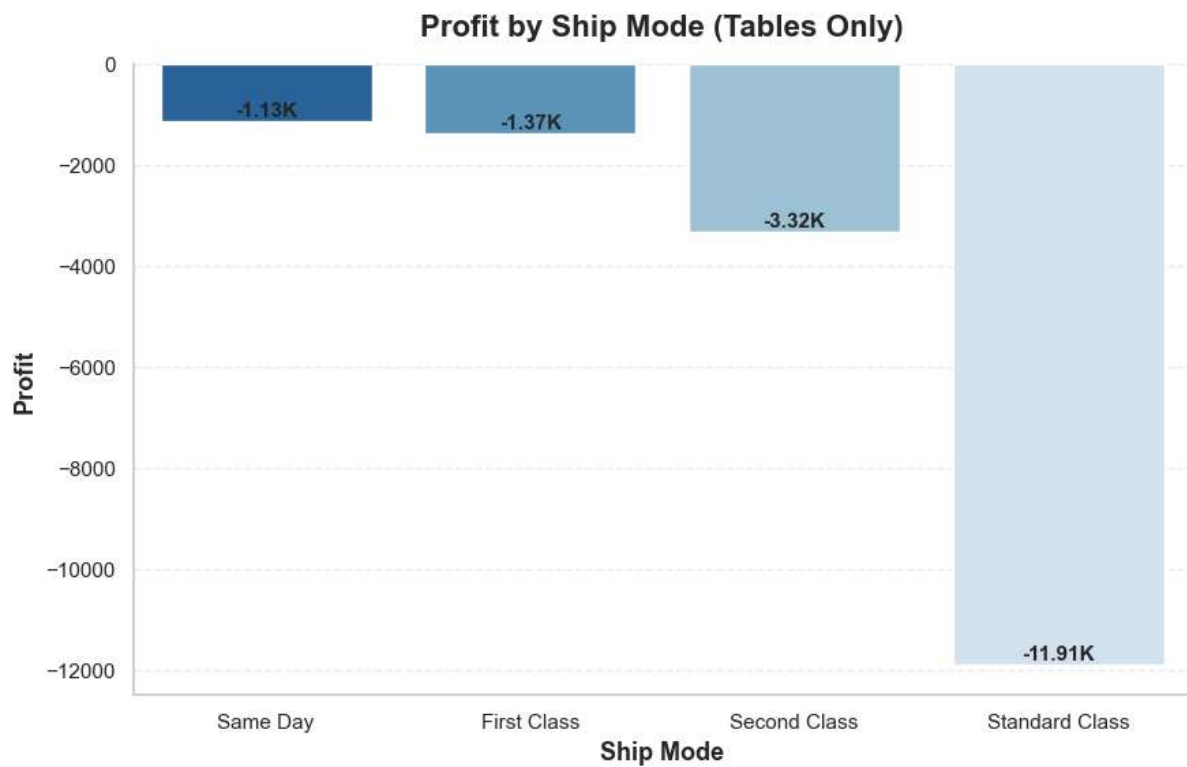
Key Findings: Customer profitability is not equally distributed.

Business Insight: Develop targeted strategies for the Home Office segment.

5.7 Shipping Mode Performance Analysis

Business Question: Which shipping mode contributes the highest profitability?

Observation: Standard Class generated the highest profit whereas Same Day shipping generated the lowest.



Key Findings: Shipping mode affects profitability.

Business Insight: Review Same Day shipping costs and pricing strategy.

6. Overall Key Findings

- Technology generated the highest sales and profit.
- Furniture generated high sales but comparatively low profitability.
- Tables recorded the highest losses within Furniture.
- Higher discounts were associated with lower profit.
- West region achieved the highest profitability while East recorded the lowest.
- Home Office contributed the lowest profit among customer segments.
- Standard Class was the most profitable shipping mode.
- The dashboard provides a quick executive view of business performance.

7. Business Recommendations

- Review pricing and discount strategy for Tables.
- Analyze Furniture cost structure.
- Optimize discount policies.
- Investigate low profitability in the East region.
- Develop targeted strategies for Home Office customers.
- Review Same Day shipping efficiency.
- Use dashboards for continuous business monitoring.

8. Conclusion

The dashboard converts business data into a clear visual summary that helps identify profitability trends across categories, discounts, regions, customer segments and shipping modes. The project demonstrates how effective data visualization supports faster and more informed business decisions.

The findings demonstrate the importance of combining data visualization with business analysis to support decision-making. The dashboard can also serve as a foundation for future interactive Power BI or Tableau dashboards..